



Rudrakshi Agarwal

EDUCATION: Student **1st year B-TECH (CSE)**
IIITD (Indraprastha institute of information and technology)

GRADE: SEM1- **9.2** SEM2-(ongoing) **12th BOARD:** 93.4%(CBSE)

EXPERIENCE:

- **BYLD CLUB-** I am a member of BYLD club from **last 10 months**. BYLD is the esteemed software development club at IIITD. We delve into web development, app development and open source contributions.
- **CiPD(Centre for Intelligent Product Development)-** I am part of the team at CiPD from **last 9 months**. It is a specialized research and training hub focused on fostering innovation in intelligent electronic product development.
- **Undergraduate Research Club-** I recently joined the club. focus is on mentoring junior researchers and promoting high-quality AI/ML research through collaborative learning and research groups.

PROJECTS:

- **Delhi Metro Route and Schedule Simulator-**
A Python-based metro simulation system using real DMRC data to compute optimal routes, next train arrivals, travel time, and dynamic fares across Blue, Magenta, and Red lines with peak/off-peak scheduling and interchange handling, using basic file handling without external libraries.
- **RISC-V Assembler project.-**
RV32I assembler in Python that converts assembly instructions into 32-bit binary machine code, supporting R, I, S, B, U, and J-type instruction encoding, label resolution, PC-relative offset calculations for branches/jumps, empty-line handling, and detailed line-number-based error reporting for invalid syntax, registers, immediates, and missing halt conditions.
- **RISC-V Simulator project.-**
A Python-based RV32I simulator that executes 32-bit binary machine code with support for R, I, S, B, U, and J-type instructions, simulating registers, memory, branching, and PC updates with instruction tracing, memory dumps, alignment checks, infinite loop detection, and line-number-based runtime error handling.

PROFILE

First-year B.Tech CSE student and tech enthusiast interested in learning new technologies, exploring different computing fields, and building practical projects. Quick learner who enjoys solving problems and improving programming skills.

SKILLS

Programming: Python, C++, C (basics)

HTML, CSS, Javascript, React

Libraries (basics): NumPy, Pandas, Matplotlib

Tools: GitHub, VS Code, Figma, Canva

Concepts: Data Structures (Basic), Problem Solving, OOPS (Basic)

Soft Skills: Leadership, Problem solving, Teamwork, Communication, Time Management, Multitasking, Cooperation.

CONTACT



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<https://github.com/rudrakshia30>